

Guideline #1: AI Support - Not Replace - Clinical Judgment

Scenario: A hospital deploys an AI system that automatically diagnoses patients using imaging data and recommends treatment without physician review.

Recommendation: AI should enhance human decision-making rather than supplant it. Researchers need to make systems where doctors have the last say, especially when the stakes are high. IBM says that the goal of AI is to "augment human intelligence" and not replace it.

- Allowed: Suggesting diagnoses and risks by AI
- Not allowed: Autonomous decisions by AI without human involvement

Guideline #2: Ensure Transparency & Explainability of AI Decisions

Scenario: A machine learning model predicts cancer risk, but clinicians cannot explain how the model reached its conclusion.

Recommendation: Researchers need to make sure that AI systems are clear and easy to understand so that doctors and patients can understand the choices made. Not being open makes people less trusting and stops them from using it in care settings.

- Provide interpretable outputs (e.g., features influencing diagnosis)
- Clearly disclose when AI is used in care

This directly addresses concerns in [2] about “lack of transparency” and clinician trust.

Guideline #3: Protect Patient Autonomy (Right to Opt Out of AI Use)

Scenario: A healthcare system integrates AI into all diagnostic workflows, leaving patients no option to receive non-AI-based care.

Recommendation: Patients should have real freedom, which includes the choice to opt out of AI-assisted decisions when possible. Ethical frameworks stress the importance of respecting autonomy and informed consent, which includes letting patients know when AI is involved.

Researchers must design systems that:

- Inform patients about AI involvement
- Provide alternatives when possible

Furthermore, patients may not have the ability to opt out completely (e.g., system-wide tools), but choice should be minimized as much as possible for coercion.

Guideline #4: Establish Clear Accountability for AI Outcomes

Scenario: An AI system recommends a treatment that leads to patient harm. It is unclear whether the developer, hospital, or physician is responsible.

Recommendation: Researchers must define clear accountability structures before deployment. Ethical AI requires governance frameworks that assign responsibility for errors and outcomes.

Define:

Who validates AI outputs (clinician responsibility) & Who is liable for system errors (developers/institutions)

What are the escalation procedures for disagreements between the AI and the clinician

This directly answers the issue raised in [2] regarding “who should hold accountability.”

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